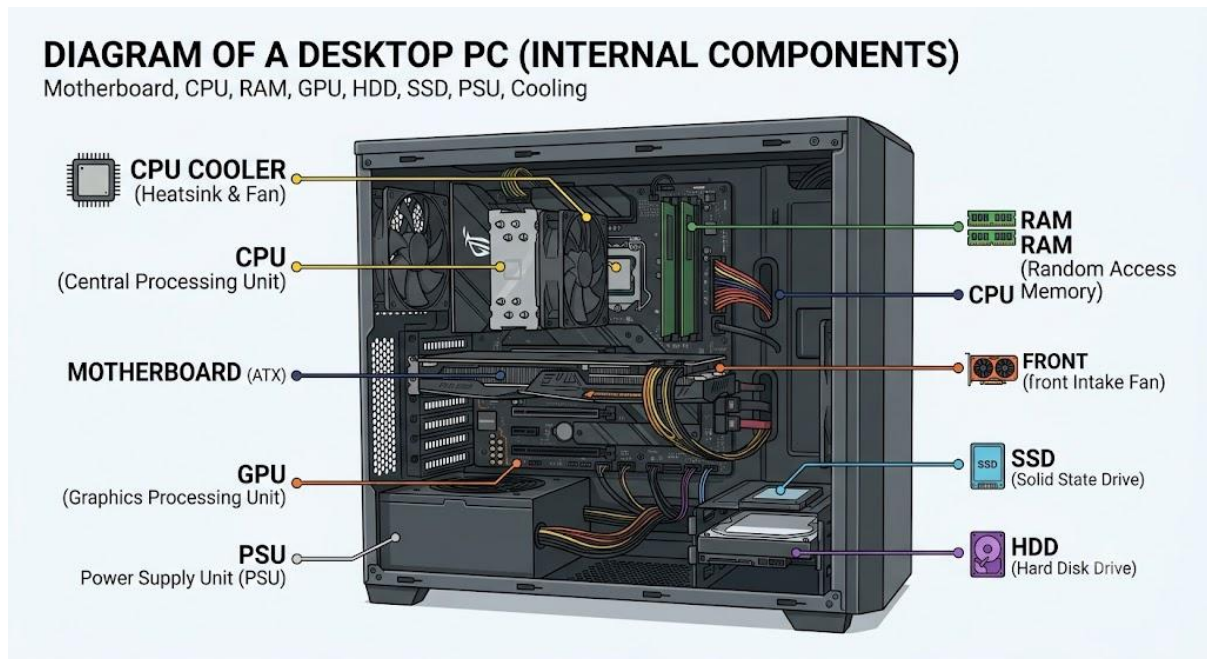


1

Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2.

Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

	HDD	SSD
	Hard Disk Drive	Solid State Drive
Speed	Slow The data transfer rate is typically around 80 to 160 MB/s. Windows boot-up and opening apps take time.	Very Fast SATA SSDs deliver 550 MB/s and modern NVMe M.2 SSDs deliver lightning-fast speeds of 3,500 to 7,000+ MB/s. The system boots almost instantly.
Durability	Low (Prone to Damage) Due to mechanical moving parts, if a laptop falls while running or experiences a major shock, there is a very high risk of data loss or hardware failure.	High (Shock-Resistant) With no moving parts, it can easily withstand physical drops, vibrations, and bumps. This makes it ideal for travel and portable devices.
Typical use	1. Bulk Storage / Backups: For long-term storage of old photos, movies, and large files. 2. Budget PCs / NAS: Where large space is needed on a tight budget.	1. Gaming PCs & Consoles: To virtually eliminate game loading times and ensure a smooth experience. 2. Modern Laptops: For slim designs, better battery life, and durability.

3.

Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

□ **CPU (Central Processing Unit):** The CPU acts as the brain of the game, handling all the background logic, physics calculations, and AI behaviors, such as calculating bullet trajectories or tracking enemy player movements on the map. It also processes your controller or keyboard inputs and tells the other hardware components what to do next.

□ **RAM (Random Access Memory):** RAM serves as the ultra-fast temporary storage area where the game loads active data, such as map layouts, player skins, and audio files, directly from your SSD or HDD. This allows the CPU to access crucial game files instantly, preventing sudden mid-game stutters or massive frame drops as you move across the map.

□ **GPU (Graphics Processing Unit):** The GPU is dedicated entirely to rendering the visual elements of the game, transforming raw data from the CPU into the actual 3D images, textures, and lighting effects you see on your monitor. A powerful GPU ensures that fast-paced action looks incredibly smooth and allows you to run the game at higher resolutions and frame rates (FPS).

4.

Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

1. GPU (Graphics Processing Unit)

- **Why it matters:** FIFA games rely heavily on visual fidelity, handling advanced volumetric lighting, 3D crowd animations, hyper-realistic pitch grass, and complex player strand-based hair physics.
- **Upgrade Reason:** Upgrading to a modern mid-range GPU (like an NVIDIA RTX series or AMD Radeon RX equivalent) will allow you to bypass the baseline recommended specs (like the GTX 1660). A newer card ensures you can easily push past 60 FPS up to high-refresh targets (144Hz+) at 1080p or 1440p resolutions without visual tearing or frame drops during heavy cutscenes and replays.

2. CPU (Central Processing Unit)

- **Why it matters:** The CPU manages the heavy cloth simulation of jerseys, dynamic crowd AI, and the complex tactical *FC IQ* systems calculating player spacing and reactions in real-time.
- **Upgrade Reason:** Modern iterations of the game engine are highly demanding on multi-core processing. Upgrading from older baseline CPUs to a modern multi-core processor (such as an Intel Core i5/i7 or AMD Ryzen 5/7 from recent generations) eliminates micro-stutters and input lag, ensuring smooth gameplay logic execution while keeping up with your GPU.

3. RAM (Random Access Memory)

- **Why it matters:** While the game lists a recommended threshold of 12 GB of memory, modern systems run background tasks, operating systems, and game launchers simultaneously.
- **Upgrade Reason:** Upgrading your system memory to **16 GB or 32 GB of dual-channel RAM** ensures that your system has plenty of high-speed overhead. This prevents memory bottlenecks, ensuring texture maps and game assets load instantaneously into memory without causing sudden stutters during fast-paced counterattacks.

4. Storage (SSD Upgrade)

- **Why it matters:** The latest versions of the game take up a massive **100 GB of storage space**.
- **Upgrade Reason:** If you are currently running the game on a traditional HDD, upgrading to a **SATA or NVMe M.2 SSD** is highly recommended. An SSD will radically slash the time it takes to boot the game, navigate through the menus of Ultimate Team, and load into a match, eliminating agonizing loading screens.